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# A field evaluation of monitor placement effects in VDT users

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**Abstract**

Appropriate visual display terminal (VDT) location is a subject of ongoing debate. Generally, visual strain is associated with higher placement, and musculoskeletal strain is associated with lower placement. Seeking resolution of the debate, this paper provides a comparison of results from previous lab-based monitor placement studies to recommendations and outcomes from viewing preference and neutral posture studies. The paper then presents results from a field study that addressed two outstanding issues: Does monitor placement in a workplace elicit postures and discomfort responses similar to those seen in laboratory settings? Results showed placements in the workplace elicited postures similar to those in lab studies. Additionally, preferred VDT location generally corresponded to the location in which less neck discomfort was reported, though that trend requires further investigation. Overall, there seems to be consistent evidence to support mid-level or somewhat higher placement, as a rule-of-thumb, considering preferred gaze angle and musculoskeletal concerns. However, optimal placement may be lower for some individuals or tasks.
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## 1. Introduction

Visual discomfort and musculoskeletal discomfort, particularly in the neck and shoulders, are occupational health concerns for people who work with computers (Bergqvist and Knave, 1994; Bergqvist et al., 1995; Hünting et al., 1981). Computer monitor placement is a key facet of workstation design. Although it is a factor that has been linked to worker discomfort, disagreement seems to persist regarding optimal placement. Standards only seem to rule out extreme locations, while more specific recommendations can conflict when based exclusively on concern for visual strain or musculoskeletal strain.

### 1.1. Placement standards

The USA standard for visual display terminal (VDT) workstations, which cites early studies of user preference (Miller and Suther, 1981; Grandjean et al., 1983; Weber et al., 1984), requires only that the entire monitor be placed within an envelope of 0 to $-60^\circ$ (that is, eye level to $60^\circ$ below eye level; ANSI, 1988). ISO and Australian standards have a similar requirement (ISO, 1992; AS, 1990), although the latter also defines an optimum viewing zone from –15 to $-45^\circ$.

### 1.2. Research findings

Epidemiological research, lab-based experiments, and longer term field studies have been conducted to explore visual and musculoskeletal effects of monitor placement on computer users. In the aggregate, these studies seem to indicate some trade-off between visual strain and musculoskeletal strain with regards to optimal placement.

#### 1.2.1. Epidemiological research

Higher monitor placement has been associated with visual stress (Bergqvist and Knave, 1994; 0 to $-20^\circ$) and musculoskeletal stress (Bergqvist et al., 1995), though the latter is more often associated with lower monitor placement (Hünting et al., 1981; Grieco et al., 1982). At least two studies, however, have shown no link between

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musculoskeletal discomfort and monitor placement (Faucett and Rempel, 1994; Sauter et al., 1991).

*1.2.2. Experimentation*

Empirical placement recommendations range from 45° below the Frankfurt plane (Lie and Fostervold, 1994) to eye level (de Wall et al., 1992; Bauer and Wittig, 1998). In theory, the former reduces visual strain, the latter musculoskeletal strain. Such disparate recommendations may occur because studies generally do not investigate both areas of concern simultaneously. It follows that recommendations from such studies would minimize the targeted strain, even at the expense of the other.

*1.2.2.1. Studies of visual strain.* Benefits to a lower gaze angle include reductions in ocular surface area (Sotoyama et al., 1996) and perceived viewing exertion (vergence effort) (Menozzi et al., 1994; Tyrrell and Leibowitz, 1990; Jaschinski et al., 1998). Results from several visual preference studies are summarized in Table 1. Except for the work by Hill and Kroemer (1986), mean preferences seem to be exhibited for gaze angles about 23–34° below the eye–ear line (GA-EEL), and centered about 9–10° below horizontal (GA-h). These studies tend to report wide ranges of results, reflecting extensive individual differences in visual capabilities.

*1.2.2.2. Studies of musculoskeletal strain.* Musculoskeletal strain can be objectively assessed through measurements of posture or muscle activity, and subjectively through reports of preference or discomfort. Tables 2 and 3 present results from several lab-based studies of the effects of monitor placement on those outcomes. These studies consistently show significant increases in activity of the cervical and thoracic extensor muscles with lower VDT placement (Table 2). Increased muscle activity is a concern, because localized muscle fatigue associated with continuous activation of low force/low threshold muscle fibers is thought to be a primary contributor to development of tension neck syndrome and myofascial pain in VDT workers and others who perform semi-static tasks (Hagberg et al., 1995; Sjøgaard and Søgaard, 1998). The studies in Table 2 found discomfort tended to be greater for lower placement, though not significantly so. Apparently many subjects do not experience discomfort while working at the computer, at least during the short exposures that are typical of lab studies. Subjects seemed to favor a mid-level location over lower locations. This is consistent with the visual field

Table 1
Sample of viewing angle preference reports. Angles are relative to horizontal reference if not otherwise specified. EEL refers to the eye–ear line. FP refers to Frankfurt plane. The EEL and FP are reported to differ by about 11° (Jampel and Shi, 1992; Menozzi et al., 1994)

| Reference | VDT-based study design | Non-VDT-based study design | Viewing preference |
|---|---|---|---|
| Grandjean et al. (1983), 68 subjects | Field: work was text entry, data entry, or conversational (interactive) | | −9° to center of screen (SD = 4.5; range = +2° to −26°); 76 cm distance pref |
| Jaschinski et al. (1998), 22 subjects | Field: Telephone directory assistance office | | −8.6° to focal pt of screen (SD = 3.9; range = −3.1 to −15.3°); 78.9 cm distance pref |
| Burgess-Limerick et al. (1998), 12 subjects | Lab<br>Task: mousing game, visual focus on screen | | −3° to −22° to center of screen, with corresponding −22 to −27° angle relative to EEL |
| Hill and Kroemer (1986), 32 subjects | | Lab<br>Targets: Landholt rings and black print text on white paper | −33° to −38° to target relative to FP, at 50 cm; −24° to −31° at 100 cm |
| Menozzi et al. (1994), 74 subjects | | Lab<br>Target: red light-emitting diodes | −12.3° relative to FP (range = +10° to −25°), similar for distances 0.33, 0.5, and 1.0 m |
| Mon-Williams et al. (1999), Experiment 1, 12 subjects | | Lab<br>Target: 12 pt text subtending 16 arc/min, media not described | −26.75° relative to EEL (SD = 4.9), for 65 cm distance |
| Mon-Williams et al. (1999), Experiment 2, 6 subjects | | Target: 12 pt black text subtending 16 arc/min on white background, media not described | −34.33° relative to EEL (SD = 3.1), for 50 cm distance, −27.4° (SD = 2.1) at 100 cm |

Table 2
Sample of lab-based studies that assessed musculoskeletal strain as a function of monitor location. Summary results for EMG, discomfort, and preference assessments are presented in this table. Table 3 contains postural assessment results. All studies required subjects to focus on a screen (computer monitor or video/TV)

| Study | Task | Time on task | EMG | Discomfort | Preference |
|---|---|---|---|---|---|
| Villanueva et al. (1996); 10 subjects | Computer game with mousing | 3 min per condition | | | |
| Villanueva et al. (1997); 10 subjects | Computer game with mousing | 20 min per condition | Mean EMG increased with lowering of monitor | | |
| Burgess-Limerick et al. (1998); 12 subjects | Computer game with mousing | 5 min per condition | | | |
| Bauer and Wittig (1998); 8 subjects | Read text on VDT | 5 min per condition | Mean EMG increased with lowering of monitor | No differences | |
| Mon-Williams et al. (1999), Part 3; 12 subjects | View text on small TV, Stroop effect task | 1 min per condition, 3 repetitions | | | |
| Burgess-Limerick et al. (2000); 12 subjects | View cartoon on video monitor | 1 min per condition | | | |
| Straker and Mekhora (2000); 20 subjects | Computer game with limited keying | 20 min per condition | Mean EMG from cervical and thoracic locations was greater for the lower location | Trend towards greater discomfort with lower placement, in those S that noted a difference | 12 preferred mid,[a] 7 low, 1 had no preference |
| Sommerich et al. (accepted); 16 subjects | Discrete reading and mousing computer tasks | 30 min per task | Static and median EMG from cervical and thoracic locations increased as monitor was lowered | Trend towards less discomfort at mid-level placement | Order of preference was mid, high, low |

[a] The study's authors labelled the placement (−10°) as "high", but in this analysis of these placement studies, the location falls into the mid category.

studies cited previously (Jaschinski et al., 1998; Grandjean et al., 1983), and the monitor location study by Burgess-Limerick et al. (1998). With the goal of reducing musculoskeletal strain, these results tend to support a mid-level or higher location, rather than placement below mid-level.

Table 3 shows consistency in induced postures among the studies (postures are depicted in Fig. 1). The data also show that there is some variability in the resultant postures, even for the same researchers working with different groups of subjects in different studies. Table 3 shows that mid-level placement consistently elicited gaze angles in the preferred range (GA-EEL in Table 1), while only about half the eye-level and moderate-low placements led to preferred GA-EEL. Mid-level placement also coincided with preferred GA-h. Evaluating the remaining data in Table 3 requires knowledge of neutral body postures, if it is assumed that strain expresses a difference from neutral.

A few studies have sought to define normal (neutral) head and neck postures, though Cailliet (1991) emphasized the distinction between correct posture and those that feel normal, but are structurally faulty; Gray et al. (1966) reported marked differences of 10–15° between "comfortable" and "best" measured postures. Nevertheless, some neutral head and neck posture data are presented in Table 4. Comparing that information with data in Table 3 suggests that neutral head posture is achieved at mid-level or moderate-low placements. Neutral neck postures seem to be attainable with all placements, but each placement can also result in pronounced neck flexion relative to vertical, that is relative to gravitational forces which the neck extensor muscles work against to maintain head and neck posture. The data in Table 3 suggest, then, that low placement would result in both visual and musculoskeletal strain, while the mid-level location seems to minimize postural and visual strain. Eye level and

Table 3
Posture outcomes from a sample of lab-based studies that assessed musculoskeletal strain as a function of monitor location[a] (see Table 2 for other strain assessment outcomes: muscle activity, discomfort, and preference). Only results for comparable locations are presented. Units for posture are deg; values in parentheses are standard deviations. "∼" indicates values are estimates from graphs or merged averages from multiple groups. Refer to Fig. 1 for depiction of angles

| Study | Target location | GA-EEL | EEL-h | Head angle | Neck angle-b \| NA-v | Trunk angle |
|---|---|---|---|---|---|---|
| Villanueva et al. (1996) | 120 cm, GA-h of −0.6° (10) | −27.9 (9.4) | 27.3 (6.4) | | | 105.9 (34.3) |
| Villanueva et al. (1997) | 120 cm | | 21.3 (5.8) | | | 103.9 (10.8) |
| Burgess-Limerick et al. (1998) | High, GA-h of −2° (1) | −21 (5) | 19 (4) | 152 (3) | 117\|44 (4) | 109 (3) |
| Bauer and Wittig (1998) | GA-h of 0° | | | | | |
| Mon-Williams et al. (1999) | GA-h of 0° | ∼−20 | ∼20 | ∼150 | ∼122\|39 | ∼109 |
| Burgess-Limerick et al. (2000) | GA-h of 0° | ∼−26 | ∼27 | ∼158 | ∼127\|43<br>∼119\|41 | 100<br>110 |
| Sommerich et al. (accepted) | GA-h of 0° | ∼−29 | ∼29 | ∼155 | ∼133\|36 | 101 |
| *Eye level: range of means* | *0 to −2* | *−20 to −29* | *19–29* | *150–158* | *117–133\|36–44* | *101–109* |
| Villanueva et al. (1996) | 100 cm, GA-h of −17.2° (11.5) | −32.7 (7.2) | 15.5 (7.9) | | | 113.0 (7.5) |
| Villanueva et al. (1997) | 100 cm | | 8.9 (6.3) | | | 97.2 (6.8) |
| Burgess-Limerick et al. (1998) | Mid, GA-h of −15° (1) | −25 (5) | 9 (5) | 149 (7) | 115\|50 (4) | 105 (6) |
| Bauer and Wittig (1998) | GA-h of −17.5° | | 8.2° below 0° GA-h (3.3) | | | |
| Mon-Williams et al. (1999) | GA-h of −15° | ∼−32 | ∼16 | ∼146 | ∼121\|40 | ∼109 |
| Burgess-Limerick et al. (2000) | GA-h of −15° | ∼−33 | ∼19 | ∼151 | ∼125\|45<br>∼118\|42 | 100<br>110 |
| Straker and Mekhora (2000) | High, GA-h of −10° | ∼−22.5 | 12.5 (4.9) | 138.1 (11.3) | 133.8\|36.6 (8.3) | 99.6 (8.7) |
| Sommerich et al. (accepted) | GA-h of −17.5° | ∼−36.5 | ∼19 | ∼147 | ∼131\|38 | 101 |
| *Mid-level: range of means* | *−10 to −17.5* | *−22.5 to −36.5* | *8.9–19* | *138.1–151* | *115–133.8\|36.6–50* | *97.2–113* |

| | | | | | | |
|---|---|---|---|---|---|---|
| Villanueva et al. (1996) | 90 cm, GA-h of −27.5° (10.9) | −39.0 (8.2) | 11.5 (8.2) | | | 99.9 (12.6) |
| Burgess-Limerick et al. (1998) | Low, GA-h of −29° (2) | −30 (5) | 1 (4) | 145 (6) | 113\|54 (6) | 103 (9) |
| Mon-Williams et al. (1999) | GA-h of −30° | ∼−40 | ∼10 | ∼142.5 | ∼120\|41 | ∼109 |
| Burgess-Limerick et al. (2000) | GA-h of −30° | ∼−39 | ∼11 | ∼146 | ∼120\|50 | 100 |
| | | | | | ∼115\|45 | 110 |
| Straker and Mekhora (2000) | Low, GA-h of −30° | ∼−28.8 | −1.2 (6.5) | 133.2 (10.0) | 128.2\|44.6 (7.6) | 97.2 (8.9) |
| *Moderate-low: range of means* | *−27.5 to −30* | *−28.8 to −40* | *−1.2–11.5* | *133.2–146* | *113–128.2\|41–54* | *97.2–110* |
| Villanueva et al. (1996) | 80 cm, GH-h of −38.5° (12.2) | −43.2 (9.9) | 4.7 (9.1) | | | 99.4 (10.3) |
| Villanueva et al. (1997) | 80 cm | | −2.0 (5.6) | | | 93.3 (8.4) |
| Bauer and Wittig (1998) | GA-h of −35° | | 13.9° below 0° GA-h (4.4) | | | |
| Sommerich et al. (accepted) | GA-h of −35° | ∼−42 | ∼7 | ∼139 | ∼127\|42 | 101 |
| *Low: range of means* | *−35 to −38.5* | *−42 to −43.2* | *−2 to 7* | *139* | *127\|42* | *93.3–101* |

[a] Researchers presented posture data using various combinations of the six angles in this table. No study presented all of the angles. Where sufficient information was presented in the original paper, angles that were not originally calculated could be and are included in this table to aid comparisons among studies.

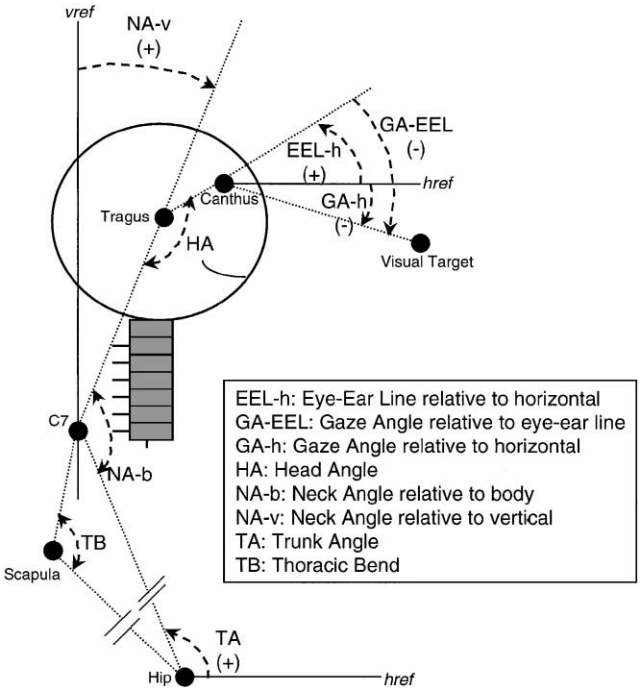



Fig. 1. Posture angles assessed in various monitor placement studies.

Table 4
Neutral posture estimates for healthy subjects. Values are in degrees, with standard deviations in parentheses. Refer to Fig. 1 for depiction of angles

| Study | Subject instruction | EEL-h | Head angle | Neck angle-b \| NA-v | Trunk angle |
|---|---|---|---|---|---|
| Mon-Williams et al. (1999) | (a) "adopt a relaxed posture" while listening to music | 16.2 (6.5) | 148 (9.5) | 120.6 \| 40.8 (6.9) | 108.6 (6.3) |
| | (b) Type while blindfolded | 12.4 (8.9) | 145.3 (6.9) | 120.8 \| 41.6 (8.4) | 107.6 (5.1) |
| Raine and Twomey (1997) | "Comfortable erect standing" | 8.9[a] | | \| 41.1 | |
| Watson and Trott (1993) | "Most comfortably balanced position" while seated | | | \| 40.9[b] | |
| Kendall and McCreary (1983) | Instruction not provided; depiction was of "ideal skeletal alignment", involving a "minimal amount of stress and strain" | ~9[c] | | \| ~34[c] | |

[a] Meatus and fold below eye defined EEL.
[b] Average NA-v was 45.5° for subjects who experienced recurring headaches.
[c] Values were estimated from diagram and photograph.

moderate-low locations each seem to provide some advantages and disadvantages.

What is lacking in this analysis is verification of the effects of monitor placement on posture in a workplace setting. An extended field study might also yield statistically significant information on visual and musculoskeletal discomfort, something that lab studies have not done. Additionally, such a study could provide preference data based on experience and context rather than first impressions. A field study was designed to address this void. Mid-level and eye level placements were studied, because those seemed likely to simultaneously induce less visual and musculoskeletal strain, based on the cited literature.

The primary questions of the study were:

(1) Do field results verify lab results, in terms of induced postures and subjective preferences, for eye level and mid-level monitor locations?
(2) Are visual and musculoskeletal discomfort outcomes different between the lab and field, given the longer exposure time in the field study?

## 2. Methods

### 2.1. Participants

Twenty software company employees volunteered for the study (8 women, 12 men; 11 software developers, 5 quality assurance analysts, 3 managers, and one technical support person). Averages for age, height, and weight were 36.6 yr ( ± 8.9), 173.3 cm ( ± 6.8), and 70.1 kg ( ± 12.6), respectively. No participants wore bifocals. All looked primarily at the VDT during work, using few or no source documents. VDT use averaged at least 6 h/day. Monitors were either 43 or 48 cm in size (17 or 19 in, for 2 and 18 subjects, respectively). Employees were invited to participate in the study if their monitors and keyboards were aligned with the mid-sagittal plane of their trunks, and if gaze angle (GA-h) as they normally worked approximated either eye level (0°) or mid-level (−17.5°). This normal GA-h would have been a function of anthropometry, workstation design, and/or personal preference, as their employer readily accommodated requests for workstation adjustments.

Participants were evenly dichotomized, based on normal GA-h. Subject grouping and experimental manipulation are depicted in Fig. 2. All participants worked with their monitors positioned normally for the first two weeks of the study. For the second two weeks, two participants in each group continued to work with their monitors positioned normally (these two subgroups of control subjects were labeled Hi–Hi and Mid–Mid), while the remaining 16 participants had their monitors relocated so that they would experience the alternate monitor location for two weeks (these two subgroups of test subjects were labeled Hi–Mid and Mid–Hi). Screen heights were changed by either adding or removing risers beneath the monitors, the typical customized intervention the employer normally provided. During data analysis, one participant from each test group was excluded because viewing angles measured during the first two weeks of the study were not consistent with screening measurements made prior to the start of the study.

### 2.2. Experimental design

A quasi-experimental, within-subjects, counterbalanced design (Campbell and Stanley, 1963) was used to test effects of monitor location on user posture, visual and musculoskeletal discomfort, and preference. Posture measures included eye–ear line relative to horizontal (EEL-h), neck angle relative to vertical (NA-v), trunk angle (TA), thoracic bending (TB), and gaze angle relative to horizontal (GA-h; see Fig. 1). These measures were directly comparable to those in Sommerich et al. (accepted). Screen tilt and distance were self-selected. Due to the large number of reports linking psychosocial work factors with visual and musculoskeletal discomfort (Bergqvist and Knave, 1994; Faucett and Rempel, 1994; Smith et al., 1981), the job content questionnaire (JCQ) (Karasek et al., 1985) was administered to participants, to account for potential confounding from work-related stress. Work organization factors, including hours worked per week, were also assessed.

#### 2.2.1. Posture assessment

A video camera was placed orthogonal to the participant's workstation at a height matching the center of rotation of the shoulder. Reflective markers were placed at the canthus, tragus, C7, inferior wing of the scapula, and center of rotation of the hip (Fig. 1). Marker location and camera height and distance were measured to ensure consistent placement on each filming day. No restrictions were placed on the participants; they were free to meet with visitors, answer the phone, or leave their offices as needed.

#### 2.2.2. Subjective assessments

At the end of each videotaping day, participants completed a survey that included questions on body part

| **Preparation** | **Week 1** | **Week 2** | **Week 3** | **Week 4** |
|---|---|---|---|---|
| Identify potential participants for appropriate normal (initial) GA-h: | VDT in initial location for all subjects. | VDT in initial location for all subjects. | VDT in alternate location for test subjects. | VDT in alternate location for test subjects. |
| | **GA-h** | **GA-h** | **GA-h** | **GA-h** |
| **Normal GA-h ≈ 0 deg** | | | | |
| n = 2 control subjects (Hi-Hi) | ≈ 0 deg | ≈ 0 deg | ≈ 0 deg | ≈ 0 deg |
| n = 8 test subjects (Hi-Mid) | ≈ 0 deg | ≈ 0 deg | ≈ -17.5 deg | ≈ -17.5 deg |
| **Normal GA-h ≈ -17.5 deg** | | | | |
| n = 2 control subjects (Mid-Mid) | ≈ -17.5 deg | ≈ -17.5 deg | ≈ -17.5 deg | ≈ -17.5 deg |
| n = 8 test subjects (Mid-Hi) | ≈ -17.5 deg | ≈ -17.5 deg | ≈ 0 deg | ≈ 0 deg |

Fig. 2. Timeline for field study.

discomfort, headache, visual discomfort, frequency of rest breaks, and workstation preferences. To describe body part and headache discomfort at the end of that work day, participants used the Borg CR-10 scale (Borg, 1982). Visual discomfort was indicated by presence or absence of each of seven symptoms (burning, itching, aching, watering, blurring, tired, or dry). Rest break frequency was assessed to determine any association with monitor location. This was the average length of time that the participant worked before needing a break from the computer. Workstation preferences were surveyed to identify likes and dislikes of the setup, including the monitor as located that day. On the final taping day, participants were asked to comment on advantages and disadvantages of each monitor location. They were also asked where they would choose to put the monitor (Hi or Mid), and were told that the monitor would be moved, if needed, to that preferred location, so that they could use it in that preferred location after the study was concluded.

### 2.3. Procedure

The study was conducted over four consecutive weeks. Each participant was observed and videotaped one day each week, for a period of one hour in the morning and one hour in the afternoon, for a total of eight data collection sessions (same day of the week and same a.m. and p.m. hours of the day for a given individual throughout the study; four taping sessions with the monitor in the normal location, then four in the alternate location). During the first session, the participant provided informed consent to participate, completed a questionnaire on typing expertise, health history, previous workstation changes, as well as a discomfort survey, and the JCQ. Anthropometry was also collected. During the first two weeks, participants worked with their VDTs in their normal locations (Fig. 2). After the fourth session, on Day 2 of Week 2, test subjects' monitors were moved to the alternate locations. Participants worked for the latter two weeks of the study with monitors in the alternate locations.

### 2.4. Data analysis and statistical testing

Videotape data were used for posture assessment and work sampling analysis; videotapes were sampled once per minute, providing 60 data points per subject per session (240 data points per subject per monitor location). To determine the effect of monitor location on posture, angles were only quantified when participants were seated orthogonal to the camera and looking at the monitor. Tapes were manually digitized, using a touch screen and stylus. To reduce variability, one analyst digitized all the video data.

Preliminary *t*-tests comparing morning and afternoon postural data revealed no significant difference, so these data sets were combined before performing ANOVA tests. The statistical model tested main effects of monitor Location (Hi or Mid), Day (1 or 2), and Group (Hi–Mid or Mid–Hi), as well as appropriate interactions. 50th percentile posture variables, body part discomfort, and headache discomfort were analyzed with this model. Contingency analysis was performed on rest break data and occurrence of visual discomfort symptoms. Musculoskeletal discomfort was further analyzed with the Mann–Whitney test to evaluate associations between discomfort and monitor location preference.

## 3. Results and discussion

The participants and conditions in which they normally worked were favorable for the purposes of this study. Participants were a homogenous group, all professionals, most with graduate degrees. They performed similar types of work and worked similar hours per week (35–40 h, considered full-time and average for the company). Rest break occurrence was found to be similar among participants, throughout the study. Results from the JCQ showed that participants experienced high job control and low psychosocial stress, typical of other previously studied computer programmers in the USA. This reduces the likelihood that visual or musculoskeletal strain would be due to psychosocial stress, and more likely that strain, if work-related, might be due to workstation set-up.

### 3.1. Posture

Based on work sampling analysis of the video tapes, 60–80% of the time participants were videotaped was spent viewing the computer monitor, with developers spending the greater percentage and managers the lesser. Fifty-two to seventy percent of the time that they were videotaped, participants viewed the computer while sitting orthogonal to the camera. Posture measurements were made during those conditions. Results from ANOVA testing showed a significant main effect of monitor location on EEL-h and GA-h (Table 5). Both showed significant decreases from Hi to Mid placement. Other terms in the model (group, day) did not affect posture outcomes. Control subjects' postures were similar to those of test subjects, and were consistent throughout the study. Except for thoracic bend, internal-referent measures were not part of the original analysis, but are provided here to complete the discussion of the posture data. Derived averages for GA–EEL, HA, and NA-b for the Hi location were $-29.7^\circ$, $159.7^\circ$, $128.2^\circ$, respectively; for the Mid location they were $-34.0^\circ$, $152.7^\circ$, and $125.1^\circ$, respectively.

Table 5
Results of analysis of variance for field study postural data

| Posture angle | Monitor location | Mean, deg (SD) | Type III SS | $F$ value | Significance | Effect size (difference between means, relative to SD) | 95% confidence interval for difference between means |
|---|---|---|---|---|---|---|---|
| Eye–ear line relative to horizontal (EEL-h) | Hi | 28.6 (6.3) | 1164.8064 | 51.29 | $p < 0.0001$ | 1.6 | (4.6, 13.6) |
| | Mid | 19.5 (5.3) | | | | | |
| Gaze angle relative to horizontal (GA-h) | Hi | −1.1 (2.8) | 2328.5574 | 184.81 | $p < 0.0001$ | 4.2 | (6.8, 20.0) |
| | Mid | −14.5 (3.6) | | | | | |
| Neck angle relative to vertical (NA-v) | Hi | 41.1 (7.9) | 62.7944 | 3.08 | $p < 0.1045$ | 0.2 | (0.9, 3.3) |
| | Mid | 43.2 (10.5) | | | | | |
| Thoracic bend (TB) | Hi | 137.2 (5.2) | 20.1057 | 0.97 | $p < 0.3448$ | 0.2 | (−1.2, 3.4) |
| | Mid | 136.1 (5.9) | | | | | |
| Trunk angle (TA) | Hi | 100.7 (6.4) | 13.2114 | 1.83 | $p < 0.2013$ | 0.1 | (−0.5, 2.5) |
| | Mid | 101.7 (7.2) | | | | | |

Posture outcomes in the field study were similar in significance to those in previous lab-based studies (refer to Table 3). In this and most previous studies, head postures and gaze angles were affected by monitor location, while neck and trunk postures were not. Posture outcomes were also similar in value and magnitude of change to outcomes in previous lab-based studies. GA–EEL tended towards the middle of the preferred range (Table 1) for the Hi location, and towards the lower end for the Mid location. EEL-h was closer to reported neutral postures (Table 4) for the Mid location. Similar to previous studies, this field study showed that posture changes do not fully compensate for changes in monitor location. Although average monitor location change was 13.4°, body adjustments totaled only 9.1°. Change in eye position (GA–EEL) made up the difference of 4.3°, or 32% of the monitor location change. With the exception of an 80% contribution found by Mon-Williams et al. (1999), in previous studies average percentage contributions from eye adjustment range from 30% to 50% between Hi and Mid locations (based on data in Table 3). In summary, the field study posture outcomes verified outcomes from previous lab-based investigations.

### 3.2. Discomfort

The neck received more discomfort reports than other body parts, though the low back was a close second, followed by the mid and upper back, the shoulders, and the wrists. Average intensity ratings of discomfort tended to be low, ranging from 0 to 2.3 on a scale of 0–10. The neck received the highest average discomfort ratings: 1.9 for Hi and 2.3 for Mid, and these two values were not significantly different. In fact, no analyses of body part discomfort reached significance, either assessed by part, or as a sum total of discomfort. Discomfort in control subjects either remained constant or improved slightly during the study. These findings are consistent with epidemiological research, in terms of body parts that are common sources of discomfort for VDT users. Results are also consistent with the lab-based studies, in terms of finding no statistically significant direct relationship between these monitor locations and body part discomfort.

Table 6
Comparison of original location of VDT, prior to the study and continued for the first two weeks of the study, and final preferred VDT location

| Original location | Final preference | | |
|---|---|---|---|
| | Hi | Mid | Between |
| Hi | 4 | 2 | 1 |
| Mid | 4 | 3 | 0 |

There was no difference in headache reporting between conditions. Mean intensity ratings were 0.7 and 0.5 for Hi and Mid, respectively. The quantities of visual discomfort symptoms reported for the two monitor locations were equally low. No differences between test conditions or trends within participants were found. Forty-five percent of the test subjects reported no visual discomfort across all four test days, as did three of the four controls. Jaschinski et al. (1998) also found comparable mean levels of eyestrain for

Table 7
Individual responses to preferred monitor position.[a] Units in degrees

| Subject | Order | Preference | GA-h | GA-EEL[b] | EEL-h | HA[b] | NA-b[b] | NA-v | Trunk | Preference comments made at end of study |
|---|---|---|---|---|---|---|---|---|---|---|
| 1-1 | Mid/Hi | Hi | 1.1 | −19.0 | 20.1 | 162.7 | 124.9 | 52.7 | 92.5 | No reason stated, although initially she noted if she could change the placement of the monitor height, she would like it somewhat higher (1–3″) |
| 14-8 | Mid/Hi | Hi | 0.8 | −29.7 | 30.5 | 165.7 | 122.5 | 45.2 | 102.3 | Re Hi: I really like the monitor setup, I feel closer to my work,[c] it's more comfortable looking more straight ahead |
| 16-10 | Hi/Mid | Hi | −6.7 | −37.2 | 30.5 | 156.4 | 134.3 | 35.9 | 99.9 | Re Mid: No real perceived difference…will put it back to its original (noted he prefers it somewhat higher 1–3″) |
| 18-11 | Hi/Mid | Hi | −4.5 | −23.2 | 18.8 | 150.6 | 128.2 | 41.9 | 100.0 | Re Mid: need to look down at my monitor, tends to make me slouch in my chair to get to eye level |
| 20-12 | Hi/Mid | Hi | −1.3 | −33.1 | 31.8 | 164.5 | 125.0 | 42.7 | 102.4 | Re Mid: it's much too low; I'm either looking down at it or slouching so that I don't have to look down |
| 2-2 | Mid/Hi | Hi | −0.9 | −32.2 | 31.4 | 156.0 | 134.3 | 34.6 | 101.2 | Re Hi: Can't tell a difference so he will leave it until/unless it causes pain (history of hand/forearm pain; also mentioned regular neck and shoulder ache although this is not reflected by in discomfort survey data collected each videotaping day). |
| 25-14 | Mid/Hi | Hi | −2.1 | −24.3 | 22.2 | 161.0 | 130.5 | 48.9 | 90.7 | Re Hi: I like having the monitor at eye level, no eye stress and less neck pain |
| 6-3 | Hi/Mid | Hi | 1.0 | −20.3 | 21.2 | 164.5 | 125.6 | 53.3 | 91.1 | Re Mid: Too low; uncomfortable, more physical strain |
| | | *Average Hi:* | *−1.6* | *−27.4* | *25.8* | *160.2* | *128.1* | *44.4* | *97.5* | |
| | | *Stdev:* | *2.8* | *6.6* | *5.7* | *5.3* | *4.5* | *7.0* | *5.1* | |
| 12-6 | Mid/Hi | Mid | −14.1 | −33.3 | 19.2 | 145.8 | 135.7 | 36.6 | 97.7 | Doesn't like High position because "I have to look up and down a greater distance to type" (look at fingers) |
| 13-7 | Hi/Mid | Mid | −12.3 | −38.4 | 26.2 | 159.5 | 125.7 | 43.3 | 101.0 | Re Mid: I feel like I can see my screens better- have several windows open at one time (caveat- my posture seems to be bending over more) |
| 15-9 | Mid/Hi | Mid | −19.4 | −37.9 | 18.5 | 162.8 | 108.5 | 54.3 | 107.3 | Re Mid: likes original height best |
| 8-4 | Mid/Hi | Mid | −11.0 | −32.1 | 21.1 | 161.3 | 118.8 | 50.2 | 101.0 | Re Hi: I keep moving closer to the screen; prefers it much lower (> 3″) |
| 9-5 | Hi/Mid | Mid | −13.4 | −37.2 | 23.8 | 155.6 | 122.5 | 41.8 | 105.8 | Re Mid: Don't have to keep my head up so high, seems to work well for me; screen is just a little further away even though I am looking down, this feels more "level" |
| | | *Average Mid:* | *−14.0* | *−35.7* | *21.7* | *157.0* | *122.2* | *45.2* | *102.6* | |
| | | *Stdev:* | *3.2* | *2.9* | *3.2* | *6.8* | *9.9* | *7.0* | *3.9* | |

[a] Data for participant who preferred to have the monitor placed in between Hi and Mid is not included in this table.
[b] These values were not measured directly from the vidoe tapes, but were derived from the other measures at a later point in the data analysis process.
[c] The monitor distance was not changed for any subjects.

monitors located at Hi or Mid locations. However, they identified an interaction between monitor distance and viewing angle, suggesting eyestrain was greater for their subjects when working with a monitor located near and Hi.

### 3.3. Location preference

At the end of the study, participants were asked to identify a final preferred location for the monitor. Eight chose the Hi location, five chose the Mid location, and one chose in between (Table 6). This resulted in a group mean preferred GA-h of $-6.5^\circ$ ($\pm 6.7$), similar to field studies by Grandjean et al. (1983) and Jaschinski et al. (1998). Follow-up at two weeks and at two months revealed no changes to these selections.

A number of factors are likely to influence a VDT user's viewing angle preference. One hypothesis regarding preference is that users choose a viewing angle that allows them to achieve reasonably low levels of visual and musculoskeletal strain, simultaneously, while not interfering with job performance, and within the physical constraints of the workstation. Table 7 contains subjective preference comments from each participant, which, in their variety, provide some support for this hypothesis. A number of factors that could have influenced the participants in this study were examined, and are discussed below.

#### 3.3.1. Visual strain

A preference for lower targets to be placed nearer than higher targets is consistent with much of the gaze angle preference literature, though Menozzi et al. (1994) found no correlation between perceived visual exertion and viewing distance. In the current study, however, gaze angle (GA-h) did not correlate with viewing distance to the center of the screen, either at the beginning (normal location) or end of the study (preferred location). Viewing distances for the Hi and Mid locations were 56.8 cm ($\pm 7.5$) and 57.1 cm ($\pm 7.9$), respectively. Viewing distances to the top and bottom of the screen are differentially affected by screen tilt. When tilt is less than the GA-h, the bottom of the screen is further away than the top. This might be more likely to occur for a Mid (or lower) placement, in an attempt to avoid glare from overhead lighting. In the current study, however, self-selected screen tilt for the Hi location averaged $3.1^\circ$ ($\pm 4.5$) tilted back from vertical, and was $10.1^\circ$ ($\pm 5.0$) for the Mid location, both similar to their respective GA-h. Screen tilt averaged $10^\circ$ in the Mid location, both for those who preferred that location and for those who chose the Hi location.

Glare from windows or overhead lights can cause problems for VDT users. In the current study, however, there was no pattern between the presence, absence, or location of an office window and monitor placement preference. The only two test subjects who worked with overhead lighting both preferred the Hi placement, along with six of the twelve who worked without overhead lighting; one Hi–Hi control and one Mid–Mid control also worked with overhead lighting. No participants commented on the influence of lighting on their choice. In summary, then, no patterns were identified regarding the workstation factors viewing distance, screen tilt, or glare that would demonstrate a trend with regards to placement preference. It may be the case, then, that monitor placement preference in these participants reflected, in part, individual differences in viewing angle preference. This is supported by some of the subjective comments in Table 7, as well as the individual and average GA–EELs in that table.

#### 3.3.2. Musculoskeletal strain

Mann–Whitney tests were performed to examine the influence of body part discomfort on gaze angle preference. Only those body parts for which discomfort was more frequently registered were tested, and of those tested only the test of neck discomfort approached significance ($p < 0.1$). The test showed that the participant's location preference generally corresponded to the location in which he or she experienced less neck discomfort (Table 8).

With the possible exceptions of the trends in neck discomfort and GA–EEL, no group-level explanations for monitor preference appeared in this data set. This finding supports the hypothesis presented earlier, describing preference as influenced by a number of factors, particularly when no egregious conditions exist. In the presence of an egregious condition, it is likely that preference within a group would be homogenous. However, when a number of factors exercise influence, yet none dominate, a heterogeneous response is more likely to result, as in this study. Several authors have concluded that optimal monitor location is that which meets the needs of the individual user (Tyrrell and Leibowitz, 1990; Jaschinski et al., 1998; Mon-Williams et al., 1999; Burgess-Limerick et al., 1998; Sommerich et al., accepted). This conclusion suggests a need for employee training in ergonomics which includes recognition of individual responses to visual and

Table 8
Correspondence between neck discomfort and final preferred VDT location

| VDT location associated with less neck discomfort | Final preference | | |
|---|---|---|---|
| | Hi | Mid | Between |
| Hi | 4 | 1 | 0 |
| Mid | 1 | 3 | 1 |
| No difference | 3 | 1 | 0 |

musculoskeletal stressors, means for easy adjustment of workstations, and training in workstation adjustment to meet individual needs.

## 4. Conclusions

When considered jointly, this study and previous research in the areas of gaze angle preference and monitor placement support location of the center of the monitor at mid-level or somewhat above, rather than below, as a reasonable starting location, based on objective and subjective research outcomes. However, individual differences in visual capabilities (such as the use of bifocals), physical make-up, work tasks, and other workstation design elements signal that fine tuning of placement is required to meet individual needs. As such, optimal placement for some individuals may occur below mid-level.

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